

Problem Arrangement

Tags:

Time Limit: 5 s Memory Limit: 64 MB
Submission: 30 AC: 15 Score: 99.47

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Description

The 11th Zhejiang Provincial Collegiate Programming Contest is coming! As a problem setter, Edward is going to arrange the order of the problems. As we know, the arrangement will have a great effect on the result of the contest. For example, it will take more time to finish the first problem if the easiest problem hides in the middle of the problem list.

There are N problems in the contest. Certainly, it's not interesting if the problems are sorted in the order of increasing difficulty. Edward decides to arrange the problems in a different way. After a careful study, he found out that the i -th problem placed in the j -th position will add P_{ij} points of "interesting value" to the contest.

Edward wrote a program which can generate a random permutation of the problems. If the total interesting value of a permutation is larger than or equal to M points, the permutation is acceptable. Edward wants to know the expected times of generation needed to obtain the first acceptable permutation.

Input

There are multiple test cases. The first line of input contains an integer T indicating the number of test cases. For each test case:

The first line contains two integers N ($1 \leq N \leq 12$) and M ($1 \leq M \leq 500$).

The next N lines, each line contains N integers. The j -th integer in the i -th line is P_{ij} ($0 \leq P_{ij} \leq 100$).

Output

For each test case, output the expected times in the form of irreducible fraction. An irreducible fraction is a fraction in which the numerator and denominator are positive integers and have no other common divisors than 1. If it is impossible to get an acceptable permutation, output "No solution" instead.

Samples

input

Copy

```
2
3 10
2 4 1
3 2 2
4 5 3
2 6
1 3
2 4
```

output [Copy](#)

3/1
No solution

Source

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